Light source

Fluorescence microscopy requires a very powerful light source such as a Xenon or Mercury Arch Lamp. The light emitted from the Mercury Arc Lamp 10–100 times brighter than most incandescent lamps and provides light in a wide range of wavelengths from ultra-violet to the infrared. Lasers or high-power LEDs were mostly used for complex fluorescence microscopy techniques.

Filter elements

A typical fluorescence microscope consists of three filters: excitation, emission and the dichroic beam splitter.

Excitation filters: It is placed within the illumination path of a fluorescence microscope. Its purpose is to filter out all wavelength of the light source, except for the excitation range of the fluorophore in the sample or specimen of interest.

Emission filters: The emission filter is placed within the imaging path of a fluorescence microscope. Its purpose is to filter out the entire excitation range and to transmit the emission range of the fluorophore in the specimen.

Dichroic filter or beam splitter: The dichroic filter or beam splitter is placed in between the excitation filter and emission filter, at 45° angle. Its purpose is to reflect the excitation wavelength towards the fluorophore in the specimen, and to transmit the emission wavelength towards the detector.



Fluorescence is called "cold light" because it does not come from a hot source like an incandescent light bulb.

2.2.3 Working Mechanism

The specimen to be observed are stained or labeled with a fluorescent dye and then illuminated with high intensity ultra violet light from mercury arc lamp. The light passes through the exciter filter that allows only blue light to pass through. Then the blue light reaches dichroic mirror and reflected downward to the specimen. The specimen labeled with fluorescent dye absorbs blue light (shorter wavelength) and emits green light. The emitted green light goes upward and passes through dichroic mirror, reflects back blue light and allows only green light to pass the objective lens, then it reaches barrier filter which allows only green light. The filtered fluorescent emission is sent to a detector where the image can be digitized Figure 2.7.

TN_GOVT_XII_Micro_Biology_CH02.indd 18 27-02-2019 13:44:13